

$$\delta_{Hand} = 0.009 \text{ in}$$

$$\delta_{FEA} = 0.009 \text{ in}$$

$$\frac{|0.009 - 0.009|}{0.009} \cdot 100 = 0\% \text{ difference}$$

$$d = 0.25 \text{ in} \quad w = 0.75 \text{ in} \quad \frac{d}{w} = 0.333$$

$$H_t = 2.30$$

$$A_{net} = (w - d) \cdot h = (0.75 - 0.25) \cdot 0.5 = 0.250 \text{ in}^2$$

$$\sigma_{net} = \frac{F}{A_{net}} = \frac{4000}{0.250} = 16000 \text{ psi} = 1.6 \text{ ksi}$$

$$\sigma_{peak} = H_t \cdot \sigma_{net} = 2.3 \cdot 16000 = 36800 \text{ psi} = 3.68 \text{ ksi}$$

$$\sigma_{peak} = 3.68 < S_y = 40$$

$$SF = 10.87$$